

# MPPT Techniques of the Solar PV under Partial Shading

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**Abstract** - During the last decade, photovoltaic energy has become a reliable source of energy with a production rate and a constantly decreasing maintenance cost thanks to the progress of technology and the evolution of the photovoltaic market.

Literature offers many MPPT algorithms that can go near the maximum power point. The characteristics of the photovoltaic panel under partial shade include several local MPPs and one global. Classical maximum power point tracking algorithms cannot achieve overall MPP. So, this work aims to study the behavioral performances of two (02) optimization techniques. They were developed to extract the overall MPP from the partially shaded PV. Maximum power point tracking performances based on PSO and P&O are evaluated for different partial shading models based on Matlab / Simulink.

**Index Terms** - Photovoltaic system, Boost converter, P&O, PSO, Partial shading.

## I. INTRODUCTION

Solar photovoltaic energy (PV) is the most popular because of its advantages such as low operational cost and environmental friendliness [1]. PV systems mainly have two disadvantages such as low conversion efficiency and low power generation. Much research has been carried out to improve its efficiency and reliability. However, the conversion efficiency of PV cells is still relatively low and PV production is highly dependent under certain environmental and operational conditions [1, 2].

The maximum power point tracking (MPPT) controllers are now widely used due to its simple implementations and ability to track the maximum power point (MPP) by varying solar irradiation.

However, tracking of MPPs will be complicated for several peaks on the P-V characteristic curve as the radiation varies. There are several conventional methods used to monitor MPP such as P&O method, incremental conductance method, Fuzzy control, PSO, etc...[3-5].

Conventional MPPT techniques give good results under uniform illumination conditions. However, this performance is not assured during partial shading because the P-V curve includes multiple peaks. Therefore they could be trapped by the local maximum. However, the objective of this work is to extract the maximum output power of the PV module under different solar radiation. The Perturb and Observe (P&O) and Practical Optimization Swarm (PSO) are shown and compared under different atmospheric conditions and under partial shade [6,7].

This article is structured as follows: Section II presents the mathematical modeling of the photovoltaic system. Section III is devoted to the different methods that are followed here. Section IV deals with the interpretation of the results obtained and the discussion on the feasibility of the different methods presented. The last section is the conclusion of the topic.

## II. MODELISATION

The model proposed for the simulation under Matlab / Simulink of the photovoltaic system is composed of a PV generator connected via a DC/DC Boost chopper for the two (02) MPPT controllers, the implementation is in the Boost converter[8].

### A. Photovoltaic cell

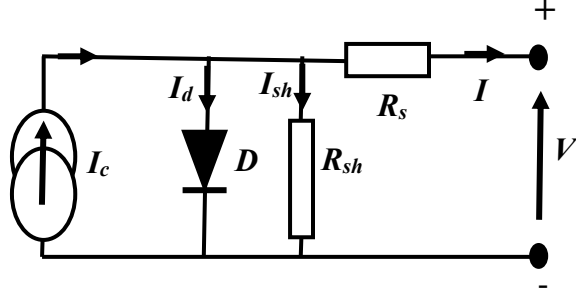


Fig. 1. PV cell equivalent electrical circuit model

Where,

$R_s(\Omega)$ : series resistance, it is the resistance offered by bulk material to flow of electrons and the resistance between the bulk material and the metal contact and it is in series with load resistance.

$R_{sh}(\Omega)$ : shunt resistance, it is the resistance caused by recombination process inside the PV cell;

$I_{cc}(A)$ : short-circuit current of the cell depending on the irradiation and the temperature;

$I(A)$ : cell current;  $V(V)$ : cell voltage;

$I_d(A)$ : diode Current and  $I_{Rsh}(A)$ : current of parallel resistor.

The solar cell equivalent circuit has been discussed in this section which is a current source connected through a diode, a shunt resistor ( $R_{sh}$ ) and a series resistor ( $R_s$ ). From The equivalent circuit we can deduce [9, 10]:

$$I = I_{cc} - I_d - I_{R_p} \quad (1)$$

The short-circuit current for any temperature ( $T$ ) is expressed by the following relation:

$$I_{cc(T)} = I_{cc(T_{ref})} \cdot (1 + k_i(T_c - T_{ref})) \quad (2)$$

$I_{cc(ref)}$ : short-circuit current for a reference temperature measured under sunshine of  $1000W/m^2$ ;

$T_{ref}$ : reference cell temperature expressed in Kelvin ( $^{\circ}K$ ) corresponding to the standard temperature  $25^{\circ}C$ ;

$T_{ref}(^{\circ}K) = 25 + 273.15^{\circ}K$ ;

$k_i$ : temperature coefficient of  $I_{ph}$  expressed in (%) ;

$I_{cc}$  is expressed by the following relation:

$$I_{cc}(E) = I_{cc}(E_0) \cdot \frac{E}{E_0} \quad (3)$$

Where,

$E_0$ : standard nominal sunshine equal to  $1000W/m^2$ .

Eq (1) can be written:

$$I = I_{cc} - I_d - \frac{V + I.R_s}{R_p} \quad (4)$$

With,

$$I_d = I_0 \left( e^{\frac{q}{nkT_c}(V + I.R_s)} - 1 \right) \quad (5)$$

Considering that the photovoltaic cell is of good quality, the current-voltage equation of the cell is written:

$$I = I_{cc} - I_0 e^{\frac{q}{nkT_c}(V + I.R_s)} \quad (6)$$

With,

$I_0(A)$ : diode saturation current;

$q = (1.602 \times 10^{-19} C)$ : electron load;

$n$ : non-ideality factor of the diode junction;

$K: 1.381 \times 10^{-23} J/K$ , Boltzmann constant;

$T_c(K)$ : effective cell temperature;

$T(K) = 273 + T(^{\circ}C)$ .

### B. Boost converter

The electrical circuit of a boost converter is presented by figure 2.

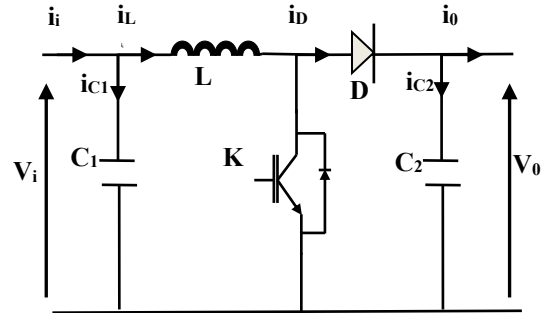


Fig. 2: Schematic diagram of a boost converter [11]

A photovoltaic generator (GPV) generates a variable power, mainly depending on irradiation and temperature.

However, it is necessary to insert a Boost converter which acts as a source-load adapter while ensuring, thanks to an adequate control strategy, the transfer of the maximum possible power to the maximum.

The duty cycle is then expressed by the following relations

$$D = \frac{T_o}{T_c} \in [0 \ 1] \quad (7)$$

### III. MAXIMUM POWER POINT TRACKING TECHNIQUES

Maximum Power Point Tracking (MPPT) techniques associated with boost converter control, allow a photovoltaic generator to operate in such a way that it always produces maximum power even under varying weather conditions such as partial shade.

In this work, the techniques of maximum power point tracking (MPPT) used are the perturbation and observation (P&O) technique and the stochastic optimization technique, the Particle Swarm Optimization (PSO) algorithm.

#### A. Perturbation and Observation

Its principle is a voltage disturbance, the power increases, the disturbance direction is maintained. Otherwise, it is reversed to resume convergence to the new MPPT. Figure 3 shows the flowchart of the P&O type MPPT control to extract maximum power at all times [12].

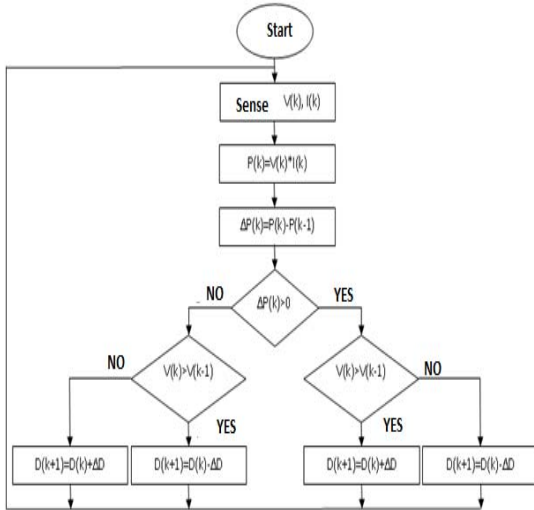


Fig. 4. P&O Algorithm

#### B. Particle Swarm Optimization

Its algorithm is inspired by the behavior of animals such as birds. This technique is based on the rules

of progressive displacement of the random positions of these particles to reach an optimal local position [13-15]. In this technique, simple rules are designed to manage particle displacement so that they gradually converge towards an overall minimum. The particle swarm optimization is based on smart swarm.

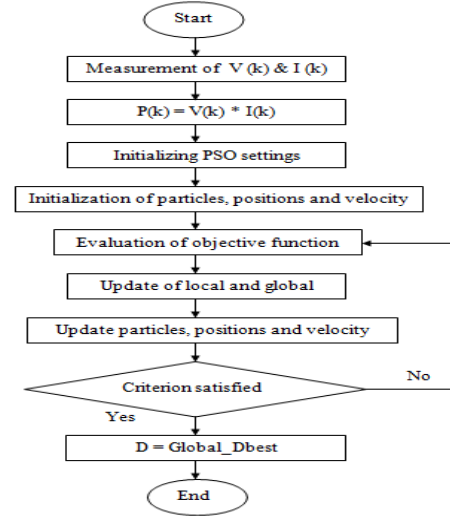


Figure 5. PSO Algorithm [12]

A flowchart of the conventional PSO algorithm is shown in Figure 5. This algorithm is executed by the following equations:

$$V_i(t+1) = w * (V_i(t) + c_1 * rand_1 * (PBest_i(t) - D_{fitness_i(t)}) + c_2 * rand_2 * (GBest_i(t) - P_i(t))) \quad (8)$$

With,

$$P_i(t+1) = P_i(t) + V_i(t+1) \quad (9)$$

And

$$D = PSO(V, I) \quad (10)$$

With,

$P$ : gear position;  $V$ : speed;  $P_{Best}$ : best position of the particle that matches Local\_DBest;  $G_{Best}$ : best position of the particle group that corresponds to Global\_DBest;  $rand$ : random variable;  $D$ : duty cycle;  $c_1$ : weight of local information;  $c_2$ : weight of global information;  $w$ : weight of inertia.

#### IV. RESULT AND DISCUSSION

Suggested PV system includes MPPT System, Step Up Converter and Photovoltaic Generator modeled and implemented in Matlab / Simulink software. For two different partial shade profiles, the monitoring efficiency and the overall performance of the MPPT techniques were evaluated. For the validation of the two techniques we simulated the photovoltaic system under standard conditions ( $E=1000W/m^2$  and  $T=25^\circ C$ ) and without partial shading.

The characteristic  $P_{pv} = f(V_{pv})$  under the standard conditions is represented by figure 5. It is noted that the PV develops a maximum power of 239W under a voltage of approximately 68V.

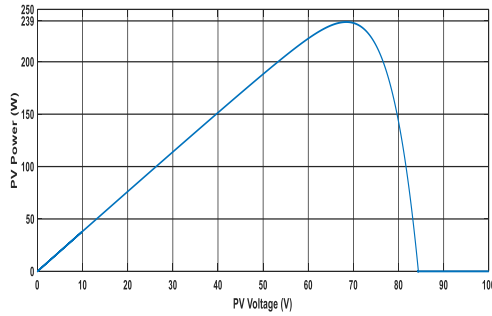


Figure 6. PV characteristic under  $E=1000W/m^2$

Under these conditions, the power trends obtained by simulations for the case of an MPPT based on the classical P&O algorithm and subsequently based on the PSO, are shown by 6 and 7, respectively. The P&O is better than the PSO in terms of speed but unlike in terms of stability on the maximum power developed.

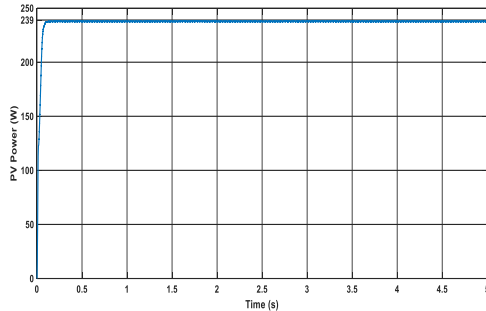


Figure 7. PV system under  $1000W/m^2$  without shading with P&O technique

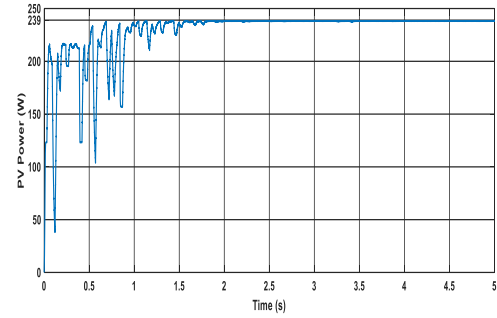


Figure 8. PV system under  $1000W/m^2$  without shading with PSO technique

Now we have imposed the conditions on the photovoltaic system in the presence of shading. In fact, two profiles were considered (See figure 8 and 9). The first presents a global maximum and three partial. And, the second, a global maximum and two partial. In addition, the simulations corresponding to the two cases are illustrated by figure 9 for technical P&O and figure 10 for technical PSO.

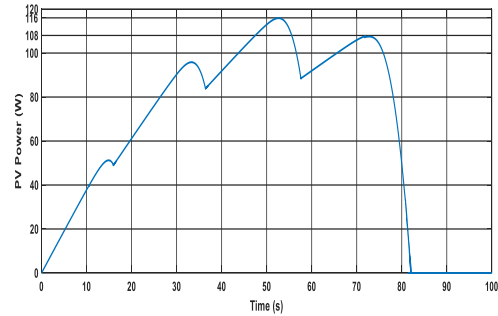


Figure9. PV characteristic under the first shading profile

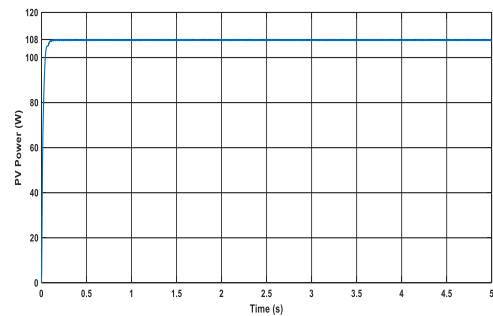


Figure 10. PV system under the first shading profile with P&O technique

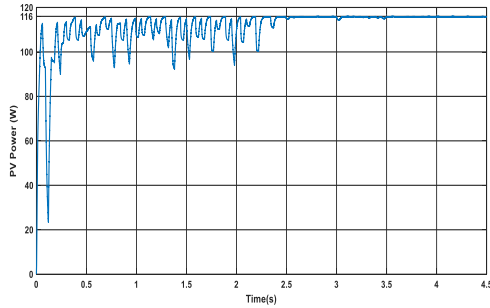


Figure 11. PV system under the first shading profile with PSO technique

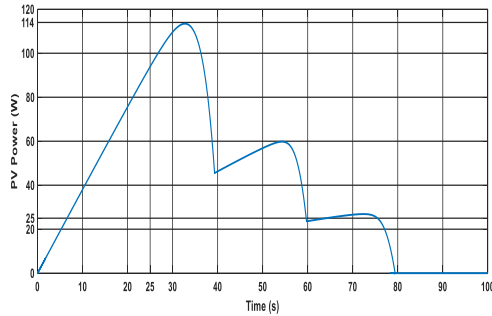


Figure 12. PV characteristic under the second shading profile

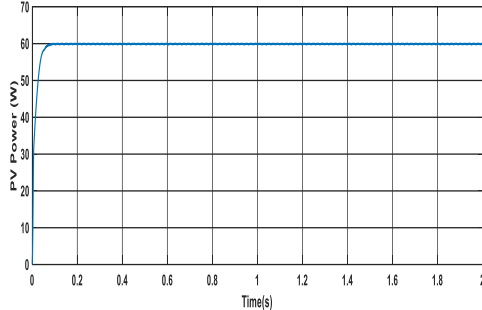


Figure 13. PV system under the second shading profile with P&O technique

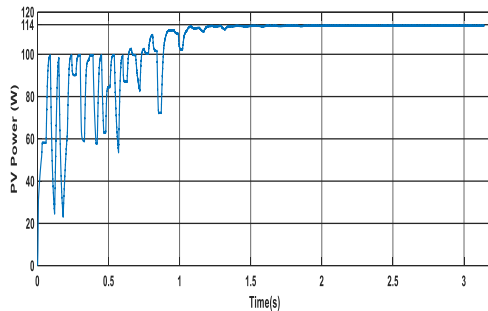


Figure 14. PV system under the second shading profile with PSO technique

The detailed results of the simulation of the PV system with different MPPT techniques show that the P&O technique failed to achieve the Global MPP and captured the first local MPP (curve  $P=f(V)$ ) due to its inability to differentiate between local and global MPP. we can deduce that the PSO technique ignores local MPP (curve  $P = f(V)$ ) and catch the global MPP.

The analysis shows that the latter method provides better results with remarkable performance.

TABLE I  
RESULTS OF PSO AND P&O MODELS

| Case N° | Irradiation (W/m <sup>2</sup> ) | MPP (W) |     |
|---------|---------------------------------|---------|-----|
|         |                                 | P&O     | PSO |
| 1       | 1000                            | 239     | 239 |
| 2       | [400 600 800 1000]              | 108     | 116 |
| 3       | [100 1000 300 1000]             | 60      | 114 |

## V. CONCLUSION

In this article, the performances of two global MPPT-based optimization algorithms are evaluated. These algorithms include P&O and PSO Techniques. The algorithms presented aim to improve the efficiency of partially shaded PVs. The evaluation of the PSO technique is carried out against the P&O technique.

The results confirm that the PSO has high precision and stability in extracting the global MPP in all cases tested, regardless of where the global MPP is compared to the conventional.

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